EDUGENIE – PROJECT DEVELOPMENT PHASES

1. Brainstorming & Ideation Phase

The brainstorming and ideation phase is the initial stage of the EduGenie project. The main objective was to identify a common problem faced by students during their learning process and develop an AI-based solution.

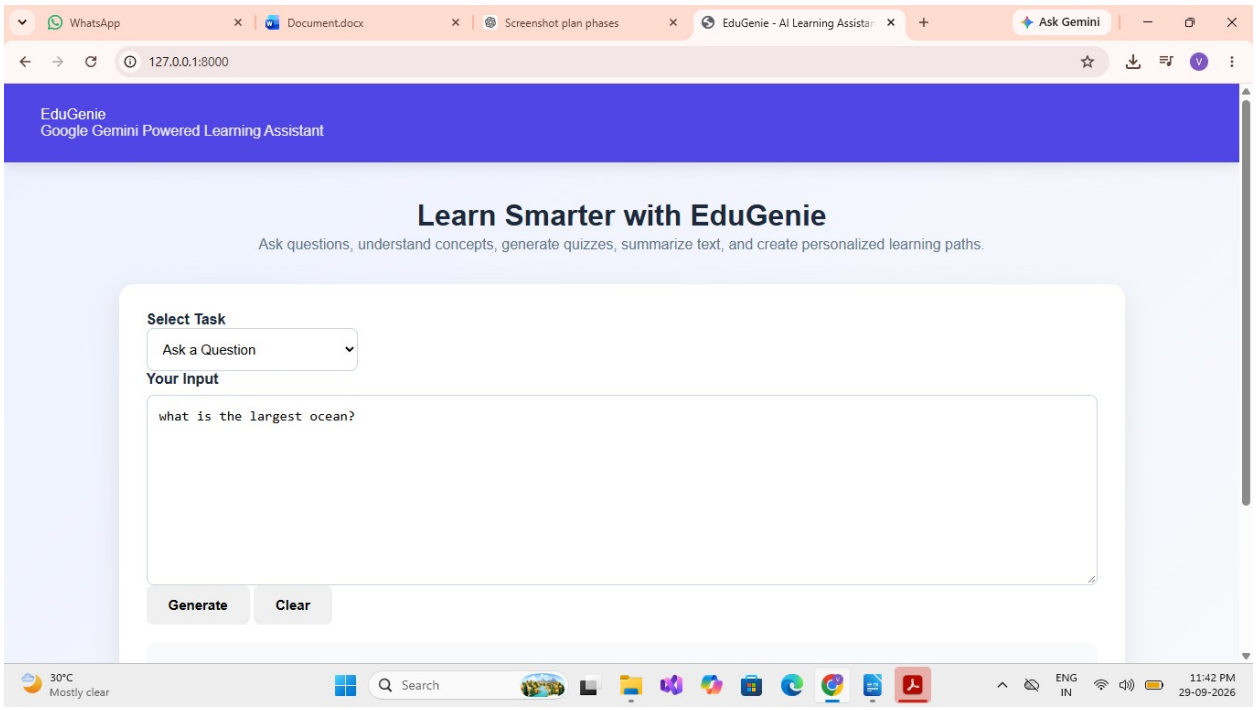
Students often find it difficult to understand complex topics, summarize lengthy study materials, prepare quizzes, and identify the correct learning resources. To address these problems, the idea of **EduGenie – Google Gemini Powered Learning Assistant** was proposed.

EduGenie is designed as a lightweight educational assistant that uses Generative AI to provide interactive learning support

Main Ideas Identified

- Provide instant answers to students' questions.
- Explain difficult concepts in simple language.
- Generate quizzes automatically.
- Summarize lengthy educational content.
- Provide personalized learning paths.
- Help students learn from beginner to advanced levels.

- Provide a simple and user-friendly interface.



Proposed Solution

The proposed solution was to develop a web-based AI learning assistant using:

- **Python** for backend development.
- **FastAPI** for REST API development.
- **HTML and CSS** for the frontend.
- **Google Gemini API** for generative AI features.
- **LaMini-Flan-T5** for lightweight concept explanation.

The idea was selected because it combines artificial intelligence with education and can be used by students with minimal technical knowledge.